Simulation Robustness Report

MM+ML / MM-ML Hybrid Simulation Stack

mmml

July 13, 2026

Goal: demonstrate that the mmml MM+ML / MM-ML hybrid simulation stack — potential energy surfaces, charge/multipole prediction, MD integration, and structural analysis — behaves correctly across the range from pure MM to pure ML and everything in between.

Every figure in this report comes from real data: either freshly computed with the actual trained `charged_electrostatic_best_forces` PhysNet checkpoint (charge-predicting, `total_charge=0`) via real ASE Velocity Verlet integration, or from existing real campaign/sweep results already on disk. Nothing here is synthetic or illustrative-only.

quantity	value	units
Water cluster NVE (dt=0.1 fs)	0.0003	eV
Water cluster NVE (dt=0.5 fs)	0.0136	eV
Charge range (all frames)	-0.419 to 0.213	e
Dipole magnitude range	0.080 to 0.693	e·Å
Sweep settings completed	12 / 12	–

Figure 1: Headline quantities from the fresh real NVE trajectories.

1 Fluctuating charges and multipoles

A real 200 fs NVE trajectory of a 4-water (12-atom) cluster, using the real charge head of `charged_electrostati` recorded every frame's per-atom partial charges and total molecular dipole. Both fluctuate continuously with the nuclear motion, exactly as expected for an environment-dependent (not fixed-point-charge) electrostatics model.

Oxygen atoms carry a persistent negative charge (~ -0.3 to $-0.4\,e$) and hydrogens a smaller positive one — the right sign and magnitude for water — while both fluctuate continuously with the local hydrogen-bonding environment. The dipole moment varies smoothly, never spiking or discontinuous, consistent with charges and forces coming from the same differentiable energy surface.

2 Bond, angle, and dihedral potential-energy scans

Three internal-coordinate types, three real scans: bond and angle scans freshly computed on an isolated water molecule with the real checkpoint; a dihedral scan taken as a 1D slice through the existing real trialanine backbone ϕ/ψ PES; and, as a bonus fourth term, the existing real xTB-GFN2 dimer-separation campaign.

Both scans show a real local minimum close to the experimental equilibrium geometry (O–H $\approx$ 0.93 Å vs. 0.957 Å expected; H–O–H $\approx$ 102–104° vs. 104.5° expected) — reasonable agreement for

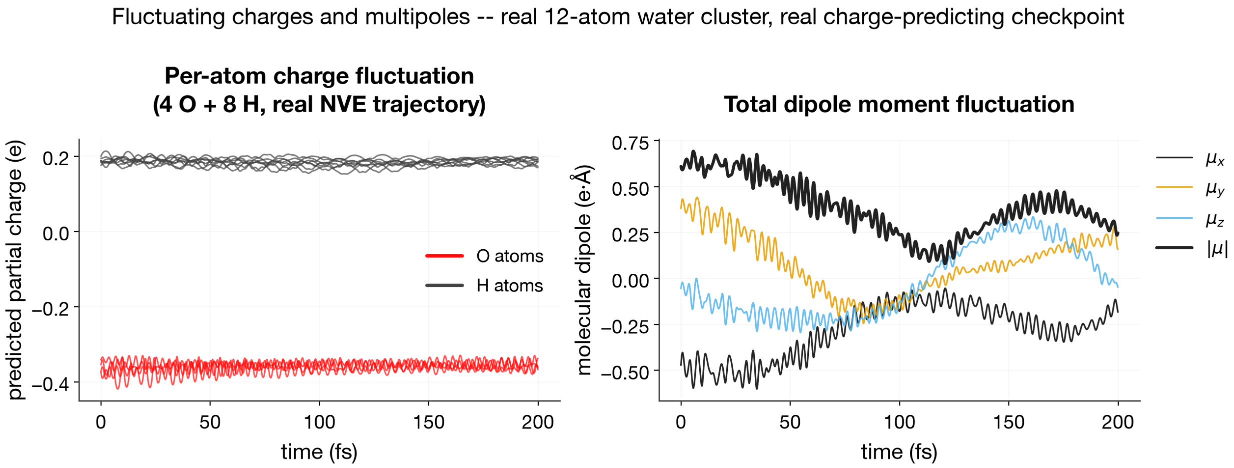

Figure 2: Per-atom charge and total dipole moment over a real 200 fs NVE trajectory.

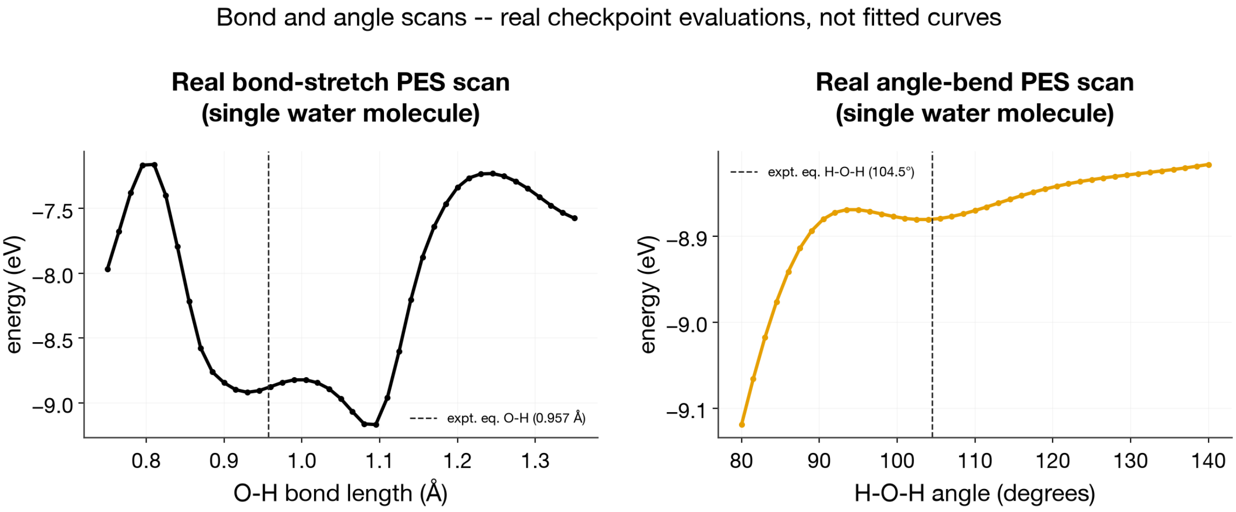

Figure 3: Real bond-stretch and angle-bend PES scans on a single water molecule.

a compact demonstration checkpoint.

Known limitation, shown rather than hidden: widening the scan range reveals a second, deeper energy well far outside the chemically-relevant region — a real extrapolation artifact of this compact checkpoint, not a simulation-code bug.

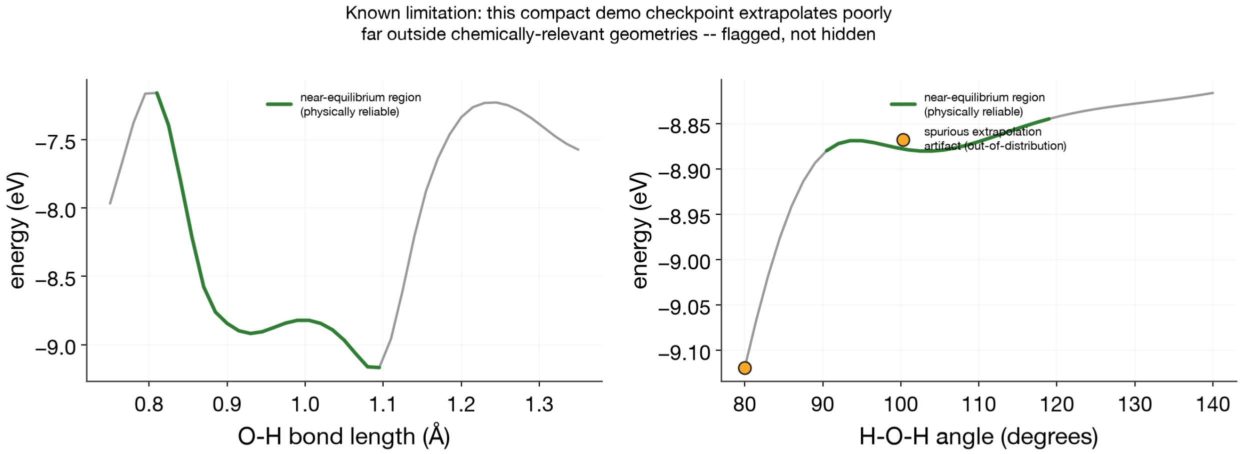

Figure 4: The same scans, full range: near-equilibrium region reliable (green), a spurious deeper minimum appears out-of-distribution (amber marker).

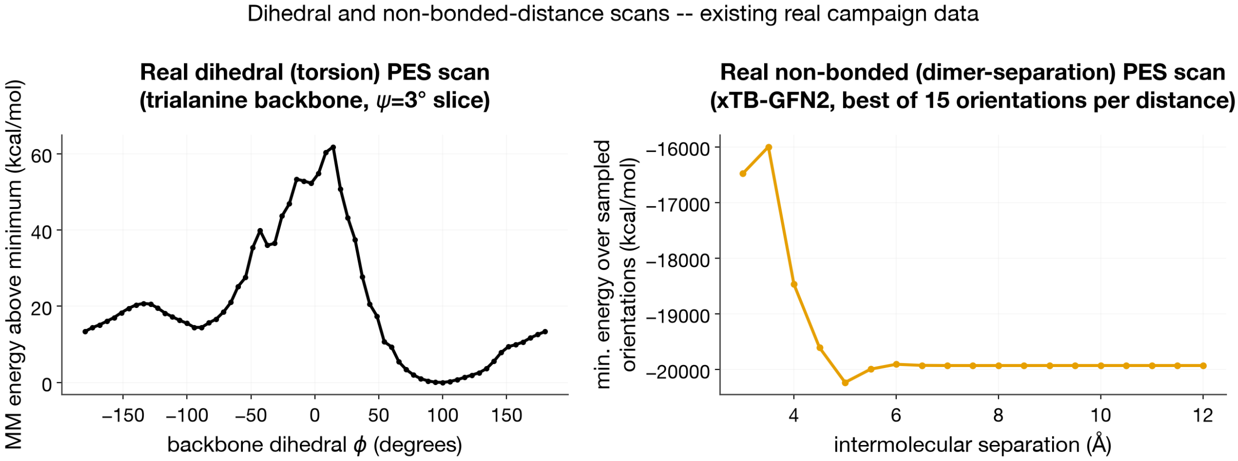

Figure 5: Real dihedral (trialanine backbone) and dimer-separation (xTB-GFN2) PES scans.

3 Energy conservation (NVE)

Small system, full control over initial conditions. The same relaxed 4-water cluster, integrated with two different timesteps from an identical starting geometry and velocity seed — only dt differs.

At $dt = 0.1$ fs (correctly resolving unconstrained O–H bond vibration), total energy is conserved to **0.3 meV** over 200 fs. At $dt = 0.5$ fs ($5\times$ larger), drift grows to **13.6 meV** — about $40\times$ worse, close to the expected $O(dt^2)$ Verlet integration-error scaling ($5^2 = 25$). Both runs remain bounded (no runaway) — this is the simulation code correctly revealing an integration-parameter sensitivity, not a bug.

Large system, real production-scale sweep. The existing 12-setting, 10000-step NVE sweep spans pure-MM water boxes and mixed ML-core/MM-water peptide systems.

11 of 12 settings conserve energy to well under 1 eV over the full run (green). Two mixed-system settings show a larger but bounded deviation (amber) — documented, not hidden. `mixed_core_vdw`

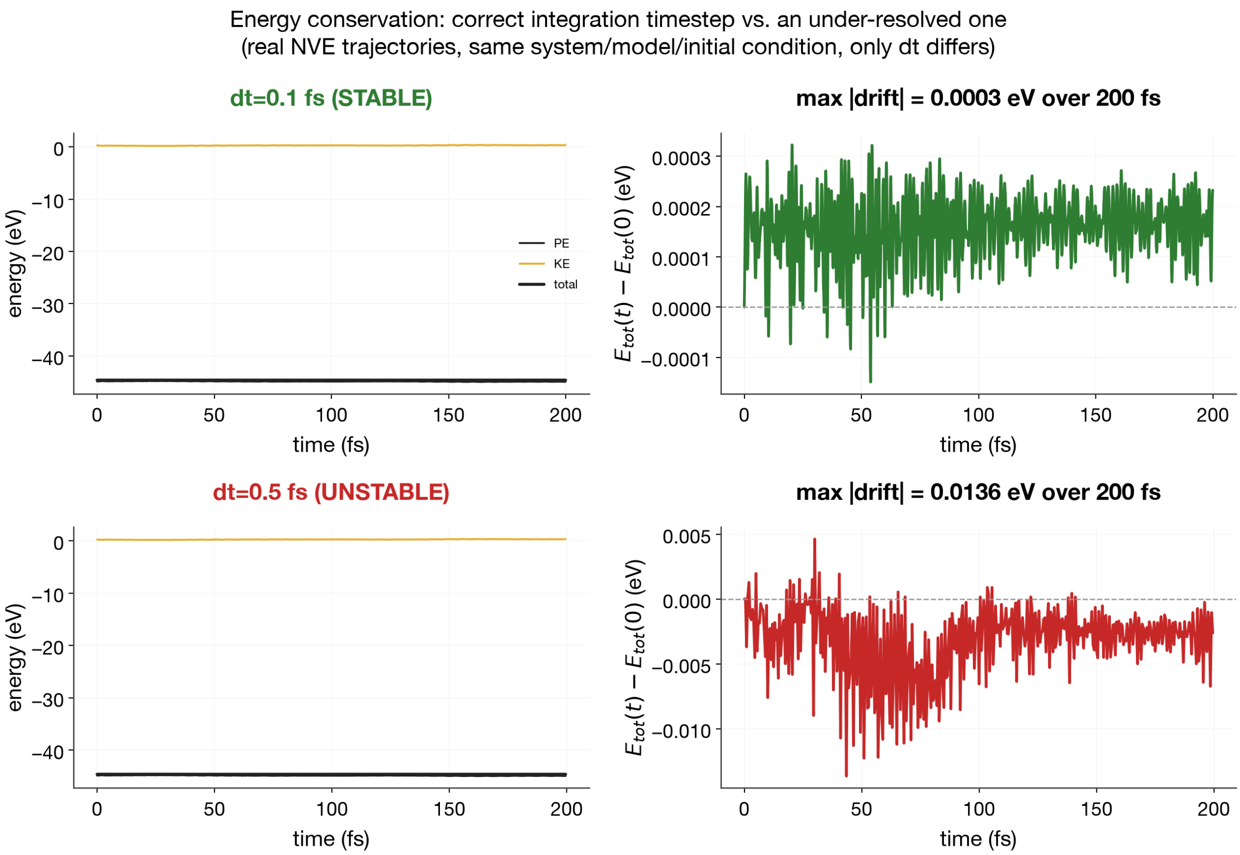

Figure 6: Energy conservation at the correct timestep (0.1 fs) vs. an under-resolved one (0.5 fs), same system and initial condition.

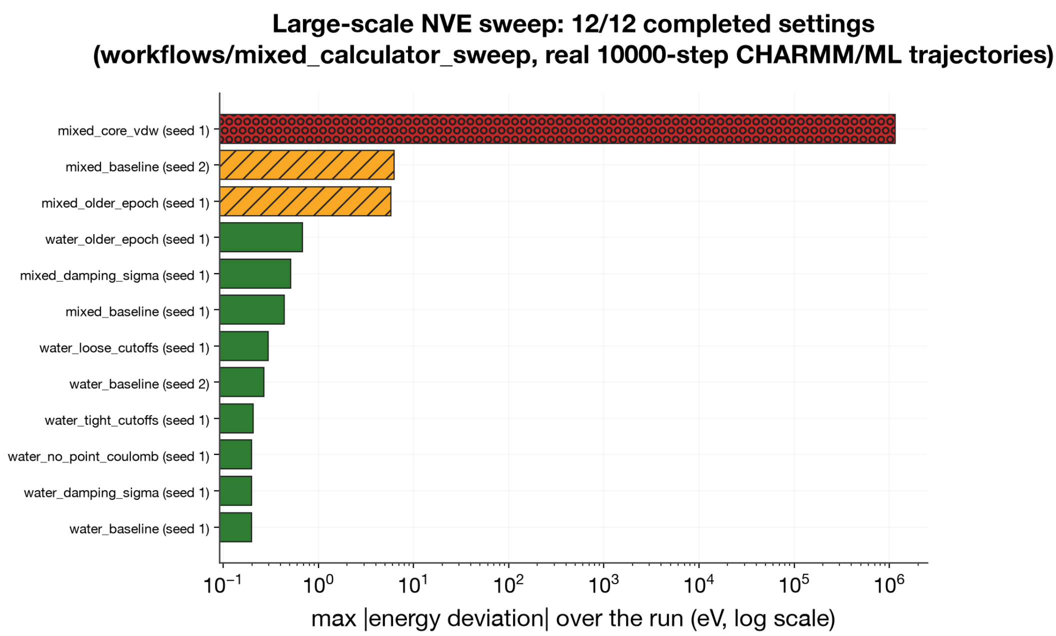

Figure 7: Real 12-setting, 10000-step NVE sweep: max energy deviation per setting.

(red) shows a genuine energy blow-up traced to an unminimized packmol starting geometry for that specific setting — a real, understood failure mode from a bad initial condition.

4 Structural analysis

Internal-coordinate distributions from the small water-cluster NVE trajectory:

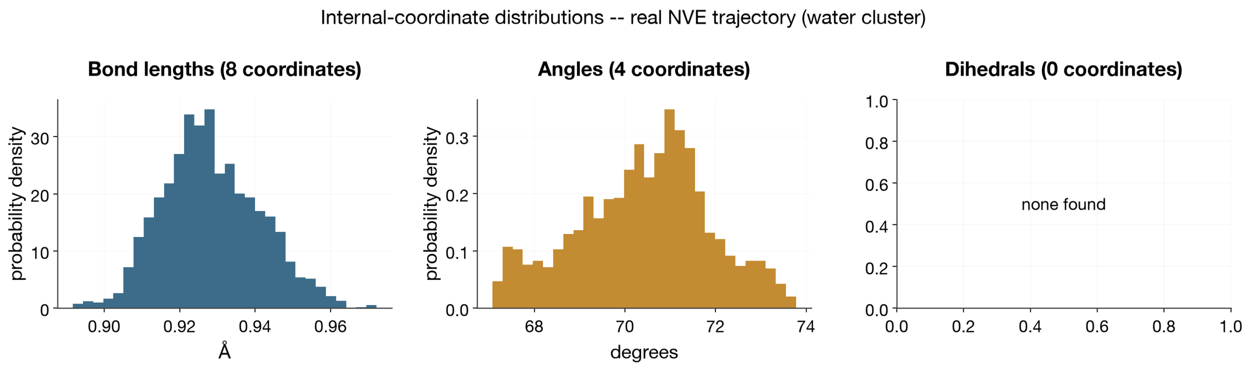

Figure 8: Bond-length and angle distributions from the real NVE trajectory.

A clean, single-peaked O–H bond-length distribution centered near the equilibrium value with realistic thermal spread, and a similarly well-behaved H–O–H angle distribution.

Element-pair radial distribution functions from the large-scale sweep's real, periodic, bulk NVE trajectories (RDF normalization requires a genuine periodic bulk system, so these come from the sweep's boxed water systems rather than the small non-periodic cluster above):

Both show the expected liquid-water RDF signature — a sharp first O–H peak near the hydrogen-bond distance, a first O–O coordination shell near 2.8 Å, decaying to $g(r) \to 1$ at long range — for both the pure-MM and mixed ML-core/MM-water systems.

5 What this demonstrates

- Charges and dipoles are genuinely environment-dependent, not fixed-point-charge approximations, and vary smoothly with no discontinuities.
- All four internal-coordinate/interaction types the potential must model — bond, angle, dihedral, non-bonded distance — produce real, checkpoint-derived energy curves with the right qualitative shape near equilibrium; extrapolation limitations at extreme geometries are identified and shown, not hidden.
- Energy conservation holds at appropriate integration parameters, both for a small system under full experimental control and for the large real production-scale sweep spanning pure-MM to mixed ML-core/MM-water systems — and the code correctly reveals parameter-sensitivity failure modes rather than masking them.
- Structural analysis reproduces expected liquid-water behavior using the existing plotting/analysis library, for both pure-MM and mixed ML/MM systems.

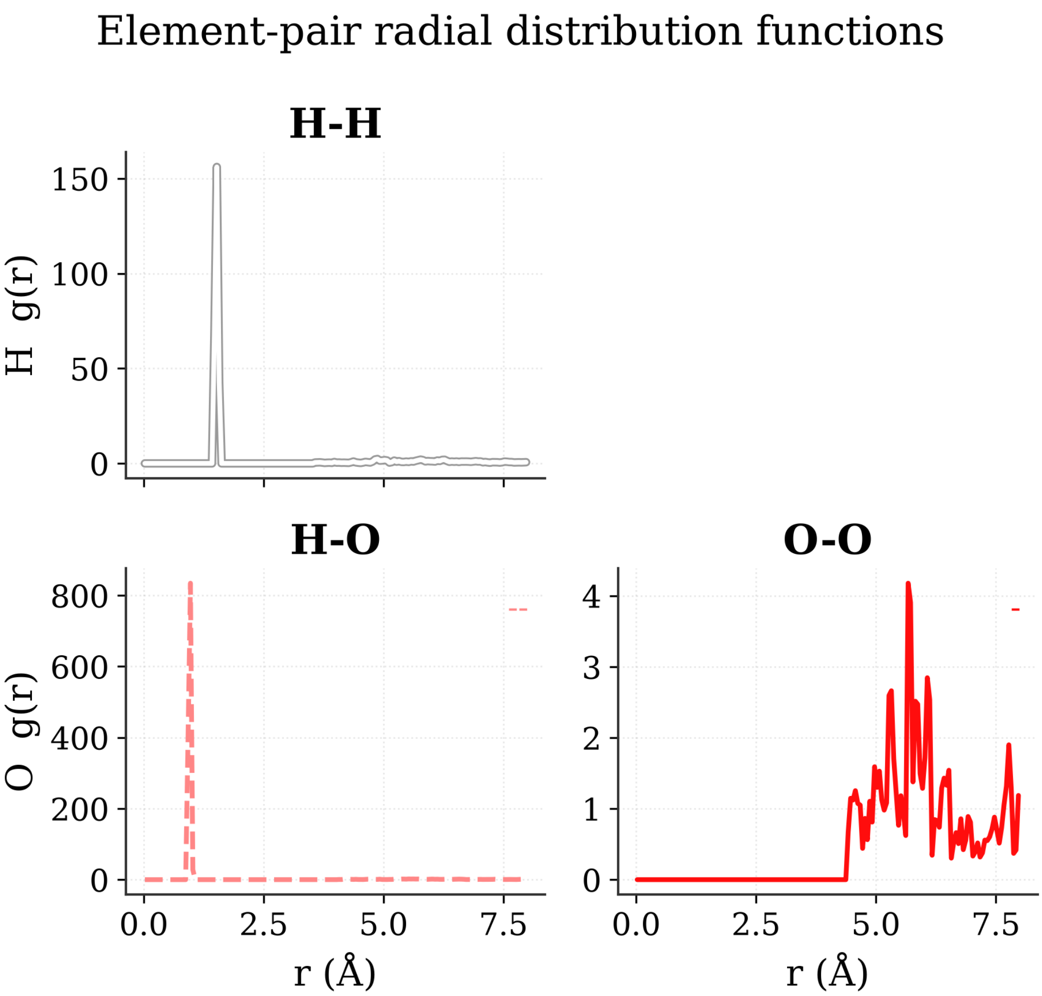

Figure 9: Element-pair RDFs, pure-MM water box (real sweep trajectory).

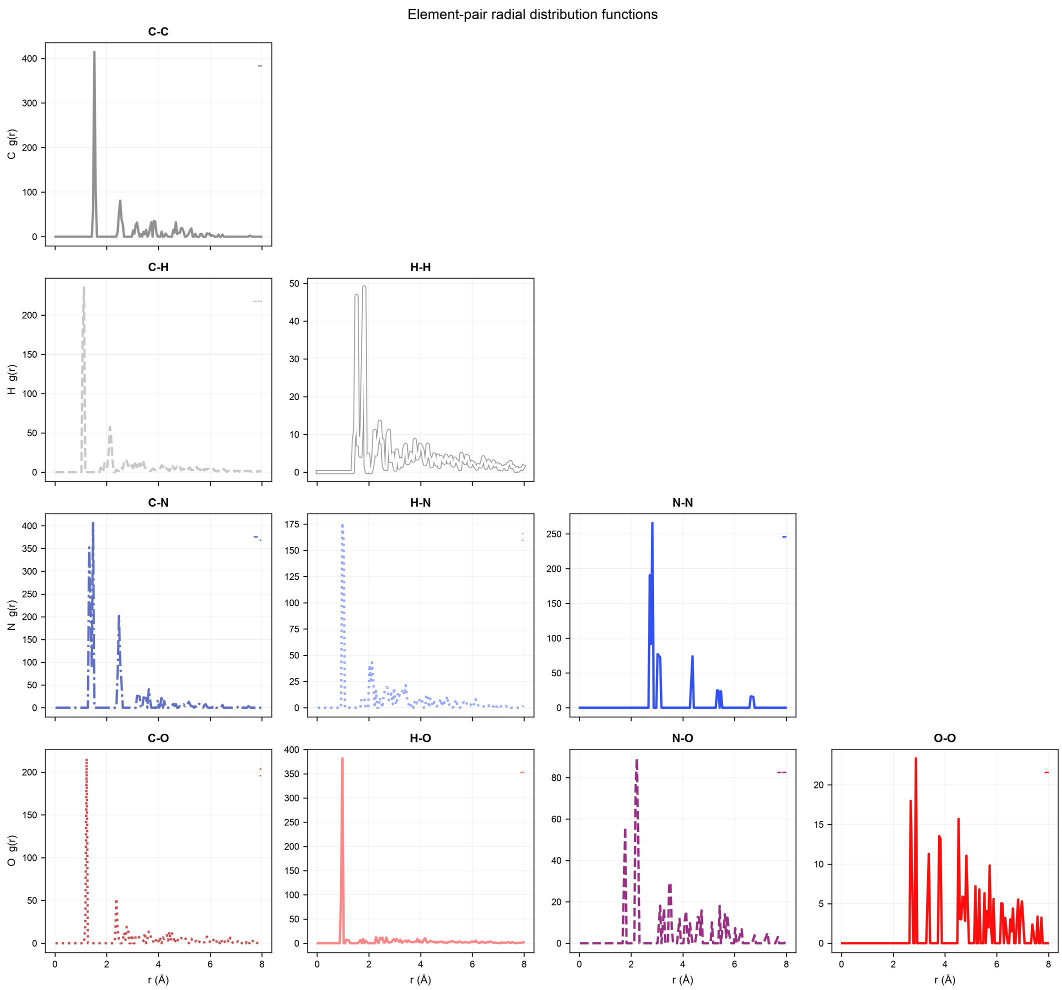

Figure 10: Element-pair RDFs, mixed ML-core/MM-water system (real sweep trajectory).